7 MATRIX REPRESENTATION OF GRAPHS

Although a pictorial representation of a graph is very convenient for a visual study, other representations are better for computer processing, A matrix is a convenient and useful way of representing a graph to a computer. Matrices lend themselves easily to mechanical manipulations. Besides, many known results of matrix algebra can be readily applied to study the structural properties of graphs from an algebraic point of view. In many applications of graph theory, such as in electrical network analysis and operations research, matrices also turn out to be the natural way of expressing the problem.

In this chapter we shall consider two most frequently used matrix representations of a graph. Also a correspondence between some graph-theoretic properties and matrix properties will be established. In view of the close tie between matrices and vector spaces, this chapter should, in fact, be looked upon as a continuation of Chapter 6. A rudimentary knowledge of matrix algebra is assumed.

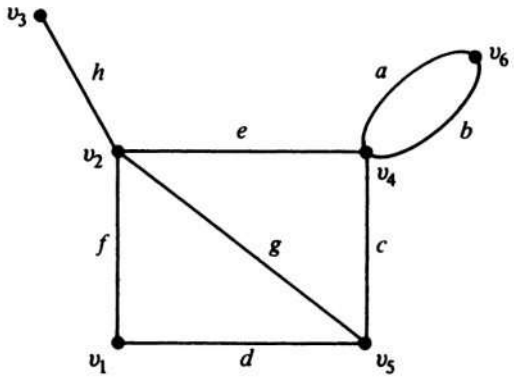
7-1. INCIDENCE MATRIX

Let G be a graph with n vertices, e edges, and no self-loops. Define an n by e matrix $A = [a_{ij}]$, whose n rows correspond to the n vertices and the e columns correspond to the e edges, as follows :

The matrix element

$a_{ij} = 1$, if jth edge e_j is incident on ith vertex v_i,and

$= 0$, otherwise.



(a)

	a	b	c	d	e	f	g	h
v_1	0	0	0	1	0	1	0	0
v_2	0	0	0	0	1	1	1	1
v_3	0	0	0	0	0	0	0	1
v_4	1	1	1	0	1	0	0	0
v_5	0	0	1	1	0	0	1	0
v_6	1	1	0	0	0	0	0	0

(b)

Fig. 7-1 Graph and its incidence matrix.

Such a matrix A is called the *vertex-edge incidence matrix*, or simply *incidence matrix*. Matrix A for a graph G is sometimes also written as A(G). A graph and its incidence matrix are shown in Fig. 7-1.

The incidence matrix contains only two elements, 0 and 1. Such a matrix is called a *binary matrix* or a (0, 1)-*matrix*. Let us stipulate that these two elements are from Galois field modulo 2.† Given any geometric representation of a graph Without self-loops, we can readily write its incidence matrix.

On the other hand, if we are given an incidence matrix A(G), we can construct its geometric graph G without ambiguity. The incidence matrix and the geometric graph contain the same information†—they are simply two alternative ways of representing the same (abstract) graph.

The following observations about the incidence matrix A can readily be made:

1. Since every edge is incident on exactly two vertices, each column of A has exactly two l′s.
2. The number of l′s in each row equals the degree of the corresponding vertex.
3. A row with all 0′s, therefore, represents an isolated vertex.
4. Parallel edges in a graph produce identical columns in its incidence matrix, for example, columns 1 and 2 in Fig. 7-1.
5. If a graph G is disconnected and consists of two components g_1 and g_2, the incidence matrix A(G) of graph G can be written in a block-diagonal form as

$$\mathbf{A}(G) = \begin{bmatrix} \mathbf{A}(g_1) & 0 \\ 0 & \mathbf{A}(g_2) \end{bmatrix}, \qquad (7\text{-}1)$$

   where $A(g_1)$ and $A(g_2)$ are the incidence matrices of components g_1 and g_2. This observation results from the fact that no edge in g_1 is incident on vertices of g_2, and vice versa. Obviously, this remark is also true for a disconnected graph with any number of components.
6. Permutation of any two rows or columns in an incidence matrix simply corresponds to relabeling the vertices and edges of the same graph. This observation leads us to Theorem 7-1.

THEOREM 7-1

Two graphs G_1 and G_2 are isomorphic if and only if their incidence matrices A(G_1) and A(G_2) differ only by permutations of rows and columns.

Rank of the Incidence Matrix: Each row in an incidence matrix A(G) may be regarded as a vector over GF(2) in the vector space of graph G. Let the vector in the first row be called A,, in the second row A_2, and so on. Thus

$$\mathbf{A}(G) = \begin{bmatrix} \mathbf{A}_1 \\ \mathbf{A}_2 \\ \cdot \\ \cdot \\ \cdot \\ \mathbf{A}_n \end{bmatrix}, \qquad (7\text{-}2)$$

Since there are exactly two 1's in every column of A, the sum of all these vectors is 0 (this being a modulo 2 sum of the corresponding entries). Thus vectors $A_1, A_2, \ldots, A_n$ are not linearly independent. Therefore, the rank of A is less than n; that is, rank A $\leq n - 1$.

Now consider the sum of any m of these n vectors ($m \leq n - 1$). If the graph is connected, A(G) cannot be partitioned, as in Eq. (7-1), such that A(g_1) is with m rows and A(g_2) with $n - m$ rows. In other words, no m by m submatrix of A(G) can be found, for $m \leq n - 1$, such that the modulo 2 sum of those m rows is equal to zero.

Since there are only two constants 0 and 1 in this field, the additions of all vectors taken m at a time for $m = 1, 2, \ldots,$ n$-$ 1 exhausts all possible linear combinations of $n - 1$ row vectors. Thus we have just shown that no linear combination of m row vectors of A (for $m \leq n - 1$) can be equal to zero. Therefore, the rank of A(G) must be at least $n - 1$.

Since the rank of A(G) is no more than $n - 1$ and is no less than $n - 1$, it must be exactly equal to $n - 1$. Hence Theorem 7-2.

THEOREM 7-2

If A(G) is an incidence matrix of a connected graph G with n vertices, the rank of A(G) is n $-$ 1.

The argument leading to Theorem 7-2 can be extended to prove that the rank of A(G) is $n - k$, if G is a disconnected graph with n vertices and k components (Problem 7-3). This is the reason why the number $n - k$ has been called the rank of a graph with k components.

If we remove any one row from the incidence matrix of a connected graph, the remaining ($n - 1$) by e submatrix is of rank $n - 1$ (Theorem 7-2). In other words, the remaining $n - 1$ row vectors are linearly independent. Thus we need only $n - 1$ rows of an incidence matrix to specify the corresponding graph completely, for $n - 1$ rows contain the same amount of information as the entire matrix. (This is obvious, since given $n - 1$ rows we can easily reconstitute the missing row, because each column in the matrix has exactly two 1's.)

Such an ($n - 1$) by e submatrix A_f of A is called a *reduced incidence matrix*. The vertex corresponding to the deleted row in A_f is called the *reference vertex*. Clearly, any vertex of a connected graph can be made the reference vertex.

Since a tree is a connected graph with n vertices and $n - 1$ edges, its reduced incidence matrix is a square matrix of order and rank $n - 1$. In other words,

COROLLARY

The reduced incidence matrix of a tree is nonsingular.

A graph with n vertices and $n - 1$ edges that is not a tree is disconnected. The rank of the incidence matrix of such a graph will be less than $n - 1$. Therefore, the $(n - 1)$ by $(n - 1)$ reduced incidence matrix of such a graph will not be nonsingular. In other words, the reduced incidence matrix of a graph is nonsingular if and only if the graph is a tree.

7-2. SUBMATRICES OF A(*G*)

Let g be a subgraph of a graph G, and let A(g) and A(G) be the incidence matrices of g and G, respectively. Clearly, A(g) is a submatrix of A(G) (possibly with rows or columns permuted). In fact, there is a one-to-one correspondence between each n by k submatrix of A(G) and a subgraph of G with k edges, k being any positive integer less than e and n being the number of vertices in G.

Submatrices of A(G) corresponding to special types of subgraphs, such as circuits, spanning trees, or cut-sets in G, will undoubtedly exhibit special properties. Theorem 7-3 gives one such property.

THEOREM 7-3

Let A(G) be an incidence matrix of a connected graph G with n vertices. An $(n - 1)$ by $(n - 1)$ submatrix of A(G) is nonsingular if and only if the $n - 1$ edges corresponding to the $n - 1$ columns of this matrix constitute a spanning tree in G.

Proof: Every square submatrix of order $n - 1$ in A(G) is the reduced incidence matrix of the same subgraph in G with $n - 1$ edges, and vice versa. From the remarks following Theorem 7-2, it is clear that a square submatrix of A(G) is nonsingular if and only if the corresponding subgraph is a tree. The tree in this case is a spanning tree, because it contains $n - 1$ edges of the n-vertex graph. Thus the theorem. ■

7-3. CIRCUIT MATRIX

Let the number of different circuits in a graph G be q and the number of edges

in G be e. Then a *circuit matrix* $B = [b_{ij}]$ of G is a q by e, (0, 1)-matrix defined as follows:

$b_{ij} = 1$, if ith circuit includes jth edge, and
$= 0$, otherwise.

To emphasize the fact that B is a circuit matrix of graph G, the circuit matrix may also be written as B(G).

The graph in Fig. 7-1(a) has four different circuits, $\{a, b\}$, $\{c, e, g\}$, $\{d, f, g\}$, and $\{c, d, f, e\}$. Therefore, its circuit matrix is a 4 by 8, (0, l)-matrix as shown:

$$B(G) = \begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \end{array} \begin{array}{c} \begin{array}{cccccccc} a & b & c & d & e & f & g & h \end{array} \\ \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 \end{bmatrix} \end{array}. \qquad (7\text{-}3)$$

The following observations can be made about a circuit matrix B(G) of a graph G:

1. A column of all zeros corresponds to a noncircuit edge (i.e., an edge that does not belong to any circuit).
2. Each row of B(G) is a circuit vector.
3. Unlike the incidence matrix, a circuit matrix is capable of representing a self-loop—the corresponding row will have a single 1.
4. The number of 1's in a row is equal to the number of edges in the corresponding circuit.
5. If graph G is separable (or disconnected) and consists of two blocks (or components) g_1 and g_2, the circuit matrix B(G) can be written in a block-diagonal form as

$$B(G) = \left[\begin{array}{c|c} B(g_1) & 0 \\ \hline 0 & B(g_2) \end{array}\right],$$

   where B(g_1) and B(g_2) are the circuit matrices of g_1 and g_2. This observation results from the fact that circuits in g_1 have no edges belonging to g_2, and vice versa (Problem 4-14).
6. Permutation of any two rows or columns in a circuit matrix simply

corresponds to relabeling the circuits and edges.

7. Two graphs G_1 and G_2 will have the same circuit matrix if and only if G_1 and G_2 are 2-isomorphic (Theorem 4-15). In other words, (unlike an incidence matrix) the circuit matrix does not specify a graph completely. It only specifies the graph within 2-isomorphism. For instance, it can be easily verified that the two graphs in Figs. 4-11 (a) and (d) have the same circuit matrix, yet the graphs are not isomorphic.

An important theorem relating the incidence matrix and the circuit matrix of a self-loop-free graph G is

THEOREM 7-4

Let B and A be, respectively, the circuit matrix and the incidence matrix (of a self-loop-free graph) whose columns are arranged using the same order of edges. Then every row of B is orthogonal to every row A; that is,

$$\mathbf{A} \cdot \mathbf{B}^T = \mathbf{B} \cdot \mathbf{A}^T = 0 \pmod 2, \tag{7-4}$$

where superscript T denotes the transposed matrix.

Proof: Consider a vertex v and a circuit Γ in the graph G. Either v is in Γ or it is not. If v is not in Γ, there is no edge in the circuit Γ that is incident on v. On the other hand, if v is in Γ, the number of those edges in the circuit Γ that are incident on v is exactly two.

With this remark in mind, consider the ith row in A and the jth row in B. Since the edges are arranged in the same order, the nonzero entries in the corresponding positions occur only if the particular edge is incident on the ith vertex and is also in the jth circuit.

If the ith vertex is not in the jth circuit, there is no such nonzero entry, and the dot product of the two rows is zero. If the ith vertex is in the jth circuit, there will be exactly two 1's in the sum of the products of individual entries. Since 1 + 1 = 0 (mod 2), the dot product of the two arbitrary rows−one from A and the other from B−is zero. Hence the theorem. ■

As an example, let us multiply the incidence matrix and transposed circuit of the graph in Fig. 7-1(a), after making sure that the edges are in the same order in both.

$$A \cdot B^T = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \text{(mod 2).}$$

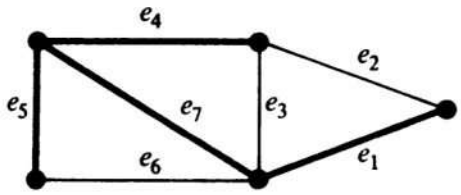
7-4. FUNDAMENTAL CIRCUIT MATRIX AND RANK OF *B*

A set of fundamental circuits (or basic circuits) with respect to any spanning tree in a connected graph, as discussed in Chapters 3 and 6, are the only independent circuits in a graph. The rest of the circuits can be obtained as ring sums (i.e., linear combinations) of these circuits. Thus, in a circuit matrix, if we retain only those rows that correspond to a set of fundamental circuits and remove all other rows, we would not lose any information. The remaining rows can be reconstituted from the rows corresponding to the set of fundamental circuits. For example, in the circuit matrix in Eq. (7-3), the fourth row is simply the mod 2 sum of the second and third rows.

A submatrix (of a circuit matrix) in which all rows correspond to a set of fundamental circuits is called a *fundamental circuit matrix* B_f. A graph and its fundamental circuit matrix with respect to a spanning tree (indicated by heavy lines) are shown in Fig. 7-2.

As in matrices A and B, permutations of rows (and/or of columns) do not affect B_f. If n is the number of vertices and e the number of edges in a connected graph, then B_f is an $(e - n + 1)$ by e matrix, because the number of fundamental circuits is $e - n + 1$, each fundamental circuit being produced by one chord.

Let us arrange the columns in B_f such that all the $e - n + 1$ chords correspond to the first $e - n + 1$ columns. Furthermore, let us rearrange the rows such that the first row corresponds to the fundamental circuit made by the chord in the first column, the second row to the fundamental circuit made by the second, and so on. This indeed is how the fundamental circuit matrix is arranged in Fig. 7-2(b).



(a)

e_2	e_3	e_6	e_1	e_4	e_5	e_7
1	0	0	1	1	0	1
0	1	0	0	1	0	1
0	0	1	0	0	1	1

(b)

Fig. 7-2 Graph and its fundamental circuit matrix (with respect to the spanning tree shown in heavy lines).

A matrix B_f thus arranged can be written as

$$B_f = [I_\mu \mid B_t], \tag{7-5}$$

where I_μ is an identity matrix of order $\mu = e - n + 1$, and B_t is the remaining μ by $(n - 1)$ submatrix, corresponding to the branches of the spanning tree.

From Eq. (7-5) it is clear that the

$$\text{rank of } B_f = \mu = e - n + 1.$$

Since B_f is a submatrix of the circuit matrix B, the

$$\text{rank of } B \geq e - n + 1.$$

In fact, we can prove Theorem 7-5.

THEOREM 7-5

If B is a circuit matrix of a connected graph G with e edges and n vertices,

$$\text{rank of B} = e - n + 1.$$

Proof: If A is an incidence matrix of G, from Eq. (7-4) we have

$$\text{A.B}^T = 0 \pmod 2.$$

Therefore, according to Sylvester's theorem (Appendix B),

$$\text{rank of A} + \text{rank of B} \leq e;$$

that is,

$$\text{rank of B} \leq e - \text{rank of A}.$$

Since $\text{rank of A} = n - 1$

we have $\text{rank of B} \leq e - n + 1.$

But $\text{rank of B} \geq e - n + 1.$

Therefore, we must have

$$\text{rank of B} = e - n + 1. \blacksquare$$

An Alternative Proof: Theorem 7-5 can also be proved by considering the circuit subspace W_Γ in the vector space W_G of a graph, as discussed in Chapter 6.

Every row in circuit matrix B is a vector in W_Γ and since the rank of any matrix is equal to the number of linearly independent rows (or columns) in the matrix, we have.

$$\text{rank of matrix B} = \text{number of linearly independent rows in B};$$

but the number of linearly independent rows in B $\leq$ number of linearly independent vectors in W_Γ, and the number of linearly independent vectors in W_Γ = dimension of $W_\Gamma = \mu$. Therefore, rank of B $\leq e - n + 1$. Since we already showed that rank of B $\geq e - n + 1$, Theorem 7-5 follows. ■

Note that in talking of spanning trees of a graph G it is necessary to assume that G is connected. In the case of a disconnected graph, we would have to consider a spanning forest and fundamental circuits with respect to this forest. It is not difficult to show (considering component by component) that if G is a disconnected graph with k components, e edges, and n vertices,

$$\text{rank of B} = \mu = e - n + k.$$

7-5. APPLICATION TO A SWITCHING NETWORK

Suppose you are given a box that contains a switching network consisting of eight switches *a, b, c, d, e, f, g,* and *h*. The switches can be turned on or off from outside. You are asked to determine how the switches are connected inside the box, without opening the box, of course.

One way to find the answer is to connect a lamp at the available terminals in series with a battery and an additional switch *k*, as shown in Fig. 7-3. And then find out which of the various combinations light up the lamp.

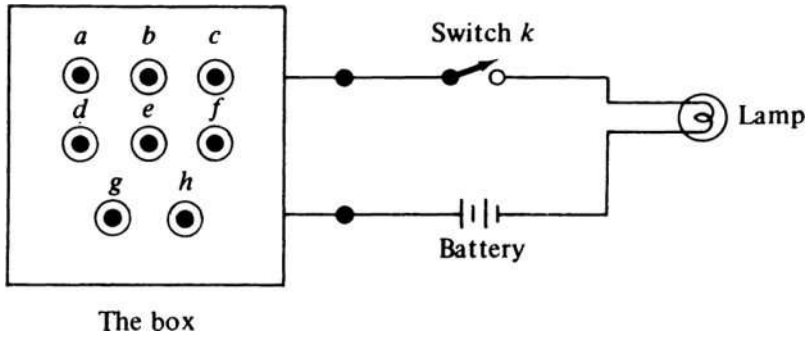


Fig. 7-3 Black box with a switching network.

In this experiment, suppose you discover that the combinations that turn on the lamp are eight:

(a, b, f, h, k), (a, b, g, k), (a, e, f, g, k), (a, e, h, k), (b, c, e, h, k), (c, f, h, k), (c, g, k), (d, k).

Solution: Consider the switching network as a graph whose edges represent switches. We can assume that the graph is connected, and has no self-loop. Since a lit lamp implies the formation of a circuit, we can regard the preceding list as a partial list of circuits in the corresponding graph. With this list we form a circuit matrix :

$$\mathbf{B} = \begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \\ 7 \\ 8 \end{array} \begin{array}{c} \begin{array}{ccccccccc} a & b & c & d & e & f & g & h & k \end{array} \\ \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 & 1 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \end{array}$$

Next, to simplify the matrix, we should remove the obviously redundant circuits. Observe that the following ring sums of circuits give rise to other circuits :

$$(a, b, g, k) \oplus (c, f, h, k) \oplus (c, g, k) = (a, b, h, k),$$
$$(a, b, g, k) \oplus (a, e, h, k) \oplus (c, g, k) = (b, c, e, h, k),$$
$$(a, e, h, k) \oplus (c, f, a, k) \oplus (c, g, k) = (a, e, f, g, k).$$

Therefore, we can delete the first, third, and fifth rows from matrix B, without any loss of information. Remaining is a 5 by 9 matrix B_1:

$$\mathbf{B_1} = \begin{array}{c} \begin{array}{ccccccccc} a & b & c & d & e & f & g & h & k \end{array} \\ \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \end{array}.$$

Our next goal is to bring matrix B_1 to the form of Eq. (7-5). For this we interchange columns to get B_2:

$$\mathbf{B}_2 = \begin{array}{c} \begin{matrix} b & e & f & g & d & a & c & h & k \end{matrix} \\ \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \end{bmatrix} \end{array}.$$

Adding the fourth row in B_2 to the first, we get B_3.

$$\mathbf{B}_3 = \begin{array}{c} \begin{matrix} b & e & f & g & d & a & c & h & k \end{matrix} \\ \left[\begin{array}{ccccc:cccc} 1 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \end{array}\right] \end{array} = [\mathbf{I}_5 \,|\, \mathbf{F}].$$

We note that there are no redundant circuits in matrix B_3, and B_3 is a fundamental circuit matrix of the required graph. Since the rank of B_3 is five, and the network was assumed to be connected, we have the following information about the graph:

number of edges $e = 9$,
nullity $\mu = 5$,
rank $r = 4$,
number of vertices $n = 5$.

Constructing a graph from its incidence matrix is simple, but constructing a graph from its fundamental circuit matrix is difficult. We shall, therefore, construct an incidence matrix from B_3.

Since the rows in the incidence matrix are orthogonal to those in B_3—according to Eq. (7-4)—we must first look for a 4 by 9 matrix M, whose rows are linearly independent and are orthogonal to those of B_3.

Since,

$$B_3 = [I_5 \,|\, F],$$

an orthogonal matrix to B_3 is

$$M = [—F^T \,¦\, I_4]$$
$$= [F^T \,¦\, I_4],$$

because in mod 2 arithmetic $-1 = 1$, [i.e., in GF(2) the additive inverse of 1 is 1].

Thus

$$M = \begin{array}{c} \begin{matrix} b & e & f & g & d & a & c & h & k \end{matrix} \\ \left[\begin{array}{ccccc:cccc} 1 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 1 \end{array}\right] \end{array}.$$

Clearly, the rank of M is four, and it is easy to check that

$$B_3 \cdot M^T = 0.$$

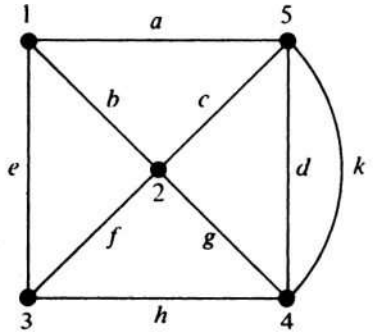
Before M can be regarded as a reduced incidence matrix, it must have at most two 1′s in each column. This can be achieved by adding (mod 2) the third row to the fourth in M, which gives us M′.

$$M' = \begin{array}{c} \begin{matrix} b & e & f & g & d & a & c & h & k \end{matrix} \\ \left[\begin{array}{ccccccccc} 1 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 \end{array}\right] \end{array}.$$

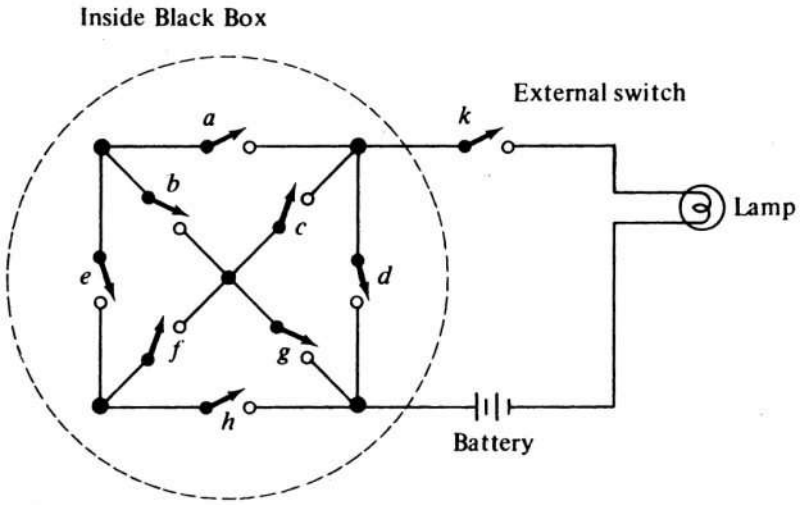
Matrix M′ is the reduced incidence matrix. The incidence matrix A can be obtained by adding a fifth row to M′ such that there are exactly two 1′ in every column; that is,

$$
\mathbf{A} = \begin{array}{c} \begin{matrix} b & e & f & g & d & a & c & h & k \end{matrix} \\ \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 0 & 1 \end{bmatrix} \end{array}.
$$

From the incidence matrix A we can readily construct the graph and hence the corresponding switching network, as shown in Fig. 7-4.



(a)



(b)

Fig. 7-4 Graph and the corresponding switching network.

7-6. CUT-SET MATRIX

Analogous to a circuit matrix, we can define a *cut-set matrix* C = $[c_{ij}]$ in which the rows correspond to the cut-sets and the columns to the edges of the graph, as follows:

$c_{ij} = 1$, if ith cut-set contains jth edge, and
$= 0$, otherwise.

For example, a graph and its cut-set matrix are shown in Fig. 7-5.

The following remarks may be made about a cut-set matrix C(G) of a graph G.

1. As in the case of the incidence matrix, a permutation of rows or columns in a cut-set matrix corresponds simply to a renaming of the cut-sets and edges, respectively.
2. Each row in C(G) is a cut-set vector.
3. A column with all 0's corresponds to an edge forming a self-loop.
4. Parallel edges produce identical columns in the cut-set matrix (e.g., first two columns in Fig. 7-5).
5. In a nonseparable graph, every set of edges incident on a vertex is a cut-set (Problem 4-8). Therefore, every row of incidence matrix A(G) is included as a row in the cut-set matrix C(G). In other words, for a nonseparable graph G, C(G) contains A(G). For a separable graph, the incidence matrix of each block is contained in the cut-set matrix. For example, the incidence matrix of the block {c, d, e, f, g} in Fig. 7-5 is the 4 by 5 submatrix of C left after deleting rows a, b, and h and columns 1, 2, 5, and 8.
6. In view of observation 5,

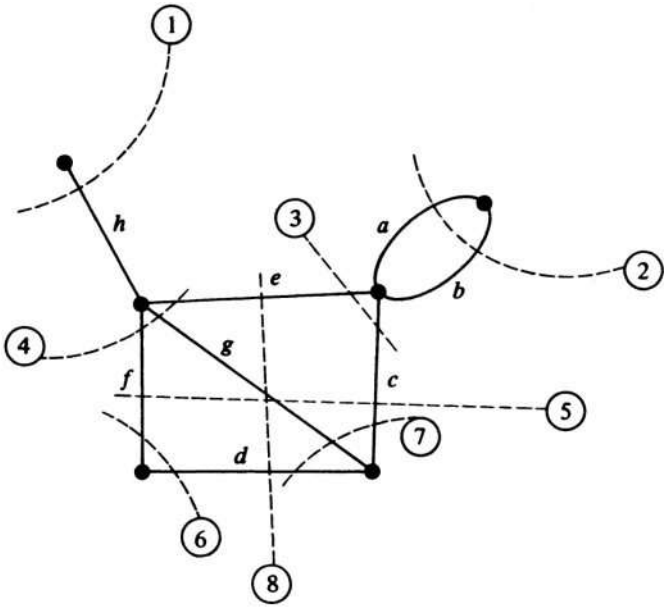
$$\text{rank of C}(G) \geq \text{rank of A}(G).$$

Hence, for a connected graph of n vertices,

$$\text{rank of C}(G) \geq n - 1. \tag{7-6}$$

7. Since the number of edges common to a cut-set and a circuit is always even, every row in C is orthogonal to every row in B, provided the edges in both B and C are arranged in the same order. In other words,

$$B \cdot C^T = C \cdot B^T = 0 \pmod 2. \tag{7-7}$$



$$C = \begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \\ 7 \\ 8 \end{array} \begin{array}{c} \begin{array}{cccccccc} a & b & c & d & e & f & g & h \end{array} \\ \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 & 0 \end{bmatrix} \end{array}$$

Fig. 7-5 Graph and its cut-set matrix.

On applying Sylvester's theorem to Eq. (7-7),

$$\text{rank of B} + \text{rank of C} \leq e.$$

and since for a connected graph

$$\text{rank of B} = e - n + 1$$

$$\text{rank of C} \leq n - 1. \tag{7-8}$$

Combining Eqs. (7-6) and (7-8),

$$\text{rank of } C = n - 1.$$

Thus we have the following important theorem for a connected graph G.

THEOREM 7-6

The rank of cut-set matrix C(G) is equal to the rank of the incidence matrix A(G), which equals the rank of graph G.

As in the case of the circuit matrix, the cut-set matrix generally has many redundant (or linearly dependent) rows. Therefore, it is convenient to define a fundamental cut-set matrix, C_f, as follows:

A fundamental cut-set matrix C_f (of a connected graph G with e edges and n vertices) is an $(n - 1)$ by e submatrix of C such that the rows correspond to the set of fundamental cut-sets with respect to some spanning tree.

As in the case of a fundamental circuit matrix, a fundamental cut-set matrix C_f can also be partitioned into two submatrices, one of which is an identity matrix I_{n-1}, of order $n - 1$. That is,

$$C_f = [C_c \mid I_{n-1}], \tag{7-9}$$

where the last $n - 1$ columns forming the identity matrix correspond to the $n - 1$ branches of the spanning tree, and the first $e - n + 1$ columns forming C_c correspond to the chords.

A connected graph and a fundamental cut-set matrix with respect to a spanning tree (shown in heavy lines) are given in Fig. 7-6.

Again note that in talking of cut-set matrices we have confined ourselves to connected graphs only. This treatment can be generalized to include disconnected graphs by considering one component at a time.

7-7. RELATIONSHIPS AMONG A_f, B_f, AND C_f

In this section we shall explore the relationships among the reduced incidence matrix A_f, the fundamental circuit matrix B_f, and the fundamental cut-set matrix C_f of a connected graph.

It has been shown that

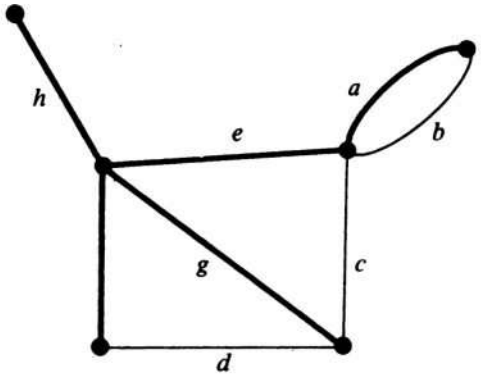
$$B_f = [I_n \mid B_t], \tag{7-5}$$

$$C_f = [C_c \mid I_{n-1}], \tag{7-9}$$

where subscript t denotes the submatrix corresponding to the branches of a spanning tree, and subscript c denotes the submatrix corresponding to the chords.

Let the spanning tree T in Eqs. (7-5) and (7-9) be the same, and let the order of the edges in both equations be the same. Furthermore, in the reduced incidence matrix A_f − of size $(n - 1)$ by e − let the edges (i.e., the columns) be arranged in the same order as in B_f and C_f. Partition A_f into two submatrices:

$$A_f = [A_c \mid A_t], \tag{7-10}$$



$$C_f = \begin{array}{c} \begin{matrix} b & c & d & a & e & f & g & h \end{matrix} \\ \left[\begin{array}{ccc:ccccc} 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{array}\right] \end{array} = [C_c \vdots I_5]$$

Fig. 7-6 Spanning tree in a graph and the corresponding fundamental cut-set matrix.

where A, consists of the $n - 1$ columns corresponding to the branches of the spanning tree T, and A_c is the remaining submatrix corresponding to the $e - n + 1$ chords.

Since the columns in A_f and B_f are arranged in the same order, from Eq. (7-4) we have (in mod 2 arithmetic)

$$A_f \cdot B_f^T = 0.$$

That is,
$$[A_c \mid A_t] \cdot \begin{bmatrix} I_\mu \\ \hline B_t^T \end{bmatrix} = 0,$$

and
$$A_c + A_t \cdot B_t^T = 0. \tag{7-11}$$

Since A_t is nonsingular, its inverse A_t^{-1} exists. Premultiplying both sides of Eq. (7-11) by A_t^{-1}, we get

$$A_t^{-1} \cdot A_c = -B_t^T. \tag{7-12}$$

Since in mod 2 arithmetic $-1 = 1$,

$$B_t^T = A_t^{-1} \cdot A_c. \tag{7-13}$$

Similarly, since the columns in B_f and C_f are arranged in the same order, according to Eq. (7-4), we have (in mod 2 arithmetic)

$$C_f \cdot B_f^T = 0.$$

That is,
$$[C_c \mid I_{n-1}] \cdot \begin{bmatrix} I_\mu \\ \hline B_t^T \end{bmatrix} = 0,$$

$$C_c = -B_t^T \tag{7-14}$$

$$= B_t^T \tag{7-15}$$

$$= A_t^{-1} \cdot A_c, \quad \text{from (7-13).}$$

For example, let us look at the following three matrices for the graph used in Figs. 7-1, 7-5, and 7-6. Using $\{a, e, f, g, h\}$ as the spanning tree, and dropping the sixth row from matrix A in Fig. 7-1 to get A_f, we have

$$\mathbf{A}_f = \begin{array}{c} \begin{matrix} b & c & d & a & e & f & g & h \end{matrix} \\ \left[\begin{array}{ccc:ccccc} 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 \end{array}\right] \end{array} = [\mathbf{A}_c \mid \mathbf{A}_t],$$

$$\mathbf{B}_f = \begin{array}{c} \begin{matrix} b & c & d & a & e & f & g & h \end{matrix} \\ \left[\begin{array}{ccc:ccccc} 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 1 & 0 \end{array}\right] \end{array} = [\mathbf{I}_3 \mid \mathbf{B}_t],$$

$$\mathbf{C}_f = \begin{array}{c} \begin{matrix} b & c & d & a & e & f & g & h \end{matrix} \\ \left[\begin{array}{ccc:ccccc} 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{array}\right] \end{array} = [\mathbf{C}_c \mid \mathbf{I}_5].$$

$\mathbf{B}_t^T = \mathbf{C}_c$ is immediate. It can also be readily verified that

$$\mathbf{A}_t^{-1} \cdot \mathbf{A}_c = \mathbf{B}_t^T.$$

This leads to three conclusions :

1. Given A or A_f, we can readily construct B_f and C_f, starting from an arbitrary spanning tree and its subgraph A, in A_f.
2. Given either B_f or C_f, we can construct the other. Thus since B_f determines a graph within 2-isomorphism, so does C_f.
3. Given either B_f or C_f, A_f in general cannot be determined completely.

7-8. PATH MATRIX

Another (0, l)-matrix often convenient to use in communication and transportation networks is the *path matrix*, A path matrix is defined for a specific pair of vertices in a graph, say (x, y), and is written as P(x,y). The rows in P(x, y) correspond to different paths between vertices x and y, and the columns correspond to the edges in G. That is, the path matrix for (x, y) vertices is P(x, y) = [p_{ij}], where

p_{ij} = 1, if jth edge lies in ith path, and
= 0, otherwise.

As an illustration, consider all paths between vertices v_3 and v_4 in Fig. 7-1(a). There are three different paths; {h, e}, {h, g, c}, and {h, f, d, c}. Let us number them 1, 2, and 3, respectively. Then we get the 3 by 8 path matrix P(v_3, v_4):

$$\mathbf{P}(v_3, v_4) = \begin{array}{c} \\ 1 \\ 2 \\ 3 \end{array} \begin{array}{c} \begin{array}{cccccccc} a & b & c & d & e & f & g & h \end{array} \\ \begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 1 & 0 & 1 \end{bmatrix} \end{array}.$$

Some of the observations one can make at once about a path matrix P(x, y) of a graph G are

1. A column of all 0's corresponds to an edge that does not lie in any path between x and y.
2. A column of all 1's corresponds to an edge that lies in every path between x and y.
3. There is no row with all 0's.
4. The ring sum of any two rows in P(x, y) corresponds to a circuit or an edge-disjoint union of circuits.

THEOREM 7-7

If the edges of a connected graph are arranged in the same order for the columns of the incidence matrix A and the path matrix P(x, y), then the product (mod 2)

$$A \cdot P^T(x, y) = M,$$

where the matrix M has l′s in two rows x and y, and the rest of the $n - 2$ rows are all 0's.

Proof: The proof is left as an exercise for the reader (Problem 7-14).

As an example, multiply the incidence matrix in Fig. 7-1 to the transposed $P(v_3, v_4)$, just discussed.

$$\mathbf{A} \cdot \mathbf{P}^T(v_3, v_4) = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

$$= \begin{array}{c} \\ v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \\ v_6 \end{array} \begin{array}{c} \begin{array}{ccc} 1 & 2 & 3 \end{array} \\ \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \end{array} \quad \text{(mod 2).}$$

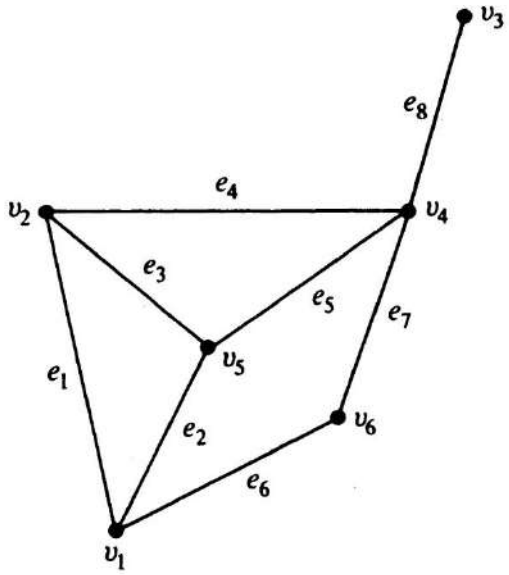
Other properties of the path matrix, such as the rank, are left for the reader to investigate on his own. It should be noted that a path matrix contains less information about the graph in general than any of the matrices A, B, or C does.

7-9. ADJACENCY MATRIX

As an alternative to the incidence matrix, it is sometimes more convenient to represent a graph by its *adjacency matrix* or *connection matrix*. The adjacency matrix of a graph G with n vertices and no parallel edges is an n by n symmetric binary matrix $X = [x_{ij}]$ defined over the ring of integers such that

$x_{ij} = 1$, if there is an edge between ith and jth vertices, and

= 0, if there is no edge between them.



$$X = \begin{array}{c} \\ v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \\ v_6 \end{array} \begin{array}{c} \begin{array}{cccccc} v_1 & v_2 & v_3 & v_4 & v_5 & v_6 \end{array} \\ \begin{bmatrix} 0 & 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 \end{bmatrix} \end{array}$$

Fig. 7-7 Simple graph and its adjacency matrix.

A simple graph and its adjacency matrix are shown in Fig. 7-7.

Observations that can be made immediately about the adjacency matrix X of a graph *G* are

1. The entries along the principal diagonal of X are all 0′s if and only if the graph has no self-loops. A self-loop at the *i*th vertex corresponds to $x_{ij} = 1$.

2. The definition of adjacency matrix makes no provision for parallel edges. This is why the adjacency matrix X was defined for graphs without parallel edges.†
3. If the graph has no self-loops (and no parallel edges, of course), the degree of a vertex equals the number of 1′s in the corresponding row or column of X.
4. Permutations of rows and of the corresponding columns imply reordering the vertices. It must be noted, however, that the rows and columns must be arranged in the same order. Thus, if two rows are interchanged in X, the corresponding columns must also be interchanged. Hence two graphs G_1, and G_2 with no parallel edges are isomorphic if and only if their adjacency matrices $X(G_t)$ and $X(G_2)$ are related:

$$X(G_2) = R^{-1} \cdot X(G_1) \cdot R,$$

where R is a permutation matrix.

5. A graph G is disconnected and is in two components g_1 and g_2 if and only if its adjacency matrix $X(G)$ can be partitioned as

$$X(G) = \begin{bmatrix} X(g_1) & 0 \\ 0 & X(g_2) \end{bmatrix},$$

where $X(g_1)$ is the adjacency matrix of the component g_1 and $X(g_2)$ is that of the component g_2.

This partitioning clearly implies that there exists no edge joining any vertex in subgraph g_1 to any vertex in subgraph g_2.

6. Given any square, symmetric, binary matrix Q of order n, one can always construct a graph G of n vertices (and no parallel edges) such that Q is the adjacency matrix of G.

Powers of X: Let us multiply by itself the 6 by 6 adjacency matrix of the simple graph in Fig. 7-7. The result, another 6 by 6 symmetric matrix X^2, is shown below (note that this is ordinary matrix multiplication in the ring of integers and *not* mod 2 multiplication) :

$$X^2 = \begin{bmatrix} 3 & 1 & 0 & 3 & 1 & 0 \\ 1 & 3 & 1 & 1 & 2 & 2 \\ 0 & 1 & 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 4 & 1 & 0 \\ 1 & 2 & 1 & 1 & 3 & 2 \\ 0 & 2 & 1 & 0 & 2 & 2 \end{bmatrix}$$

The value of an off-diagonal entry in X^2, that is, *ij*th entry (i 8800; *j)* in X^2,

= number of 1′s in the dot product of *i*th row and *j*th column (or *j*th row) of X.

= number of positions in which both *i*th and *j*th rows of X have 1′s.

= number of vertices that are adjacent to both *i*th and *j*th vertices.

= number of different paths of length two between *i*th and *j*th vertices.

Similarly, the *i*th diagonal entry in X^2 is the number of 1ʻs in the ith row (or column) of matrix X. Thus the value of each diagonal entry in X^2 equals the degree of the corresponding vertex, if the graph has no self-loops.

Since a matrix commutes with matrices that are its own power,

$$X \cdot X^2 = X^2 \cdot X = X^3.$$

And since the product of two square symmetric matrices that commute is also a symmetric matrix, X^3 is a symmetric matrix. (Again note that this is an ordinary product and not mod 2.)

The matrix X^3 for the graph of Fig. 7-7 is

$$X^3 = \begin{bmatrix} 2 & 7 & 3 & 2 & 7 & 6 \\ 7 & 4 & 1 & 8 & 5 & 2 \\ 3 & 1 & 0 & 4 & 1 & 0 \\ 2 & 8 & 4 & 2 & 8 & 7 \\ 7 & 5 & 1 & 8 & 4 & 2 \\ 6 & 2 & 0 & 7 & 2 & 0 \end{bmatrix}.$$

Let us now consider the *ij*th entry of X^3.

*ij*th entry of X^3 = dot product of *i*th row X^2 and *j*th column (or row) of X.

$= \sum_{k=1}^{n}$ *ik*th entry of $X^2 \cdot$ *kj*th entry of X.

$= \sum_{k=1}^{n}$ number all different edge squences of three edges from *i*th *j*th vertex via *k*th vertex.

= number of different edge sequences† of three edges between *i*th and *jth* vertices.

For example, consider how the 1,5th entry on X^3 for the graph of Fig. 7-7 is formed. It is given by the dot product

row 1 of X^2 · row 5 of X = (3, 1, 0, 3, 1, 0)·(l, 1, 0, 1, 0, 0)

$$= 3 + 1 + 0 + 3 + 0 + 0 = 7.$$

These seven different edge sequences of three edges between v_1 and v_5 are

$\{e_1, e_1, e_2\}$ $\{e_2, e_2, e_2\}$ $\{e_6, e_6, e_2\}$ $\{e_2, e_3, e_3\}$
$\{e_6, e_5, e_7\}$ $\{e_2, e_5, e_5\}$ $\{e_1, e_4, e_5\}$

Clearly this list includes all the paths of length three between v_1 and v_5, that is, $\{e_6, e_7, e_5\}$ and $\{e_1, e_4, e_5\}$.

It is left as an exercise for the reader to show (Problem 7-19) that the nth entry in X^3 equals twice the number of different circuits of length three (i.e., triangles) in the graph passing through the corresponding vertex v_i.

The general result that includes the properties of X, X^2, and X^3 discussed so

far is expressed in Theorem 7-8.

THEOREM 7-8

Let X be the adjacency matrix of a simple graph G. Then the ijth entry in X^r is the number of different edge sequences of r edges between vertices v_i and v_j.

Proof: The theorem holds for $r = 1$, and it has been proved for $r = 2$ and 3 also. It can be proved for any positive integer r, by induction.

In other words, assume that it holds for $r - 1$, and then evaluate the ijth entry in X, with the help of the relation

$$X^r = X^{r-1} \cdot X,$$

as was done for X^3.

The rest of the proof is left as an exercise (Problem 7-17).

COROLLARY A

In a connected graph, the distance between two vertices v_i and v_j (for $i \neq j$) is k, if and only if h is the smallest integer for which the ijth entry in x^k is nonzero.

This is a useful result in determining the distances between different pairs of vertices.

COROLLARY B

If X is the adjacency matrix of a graph G with n vertices, and

$$Y = X + X^2 + X^3 + \ldots + X^{n-1}, \qquad \text{(in the ring of integers)},$$

then G is disconnected if and only if there exists at least one entry in matrix Y that is zero.

Relationship Between A(G) *and* X(G): Recall that if a graph G has no self-loops, its incidence matrix A(G) contains all the information about G. Likewise, if G has no parallel edges, its adjacency matrix X(G) contains all the information about G. Therefore, if a graph G has neither self-loops nor parallel edges (i.e., G is a simple graph), both A(G) and X(G) contain the entire information. Thus it is natural to expect that either matrix can be obtained directly from the other, in the case of a simple graph. This relationship is given in Problem 7-23.

SUMMARY

The theory of matrices has been brought to bear upon the theory of graphs. The use of matrices in studying graphs has been amply demonstrated in this chapter.

We have seen that there are several matrices which can be associated with graphs. Two of these, the incidence matrix A and the adjacency matrix X, describe a simple graph completely, that is, up to isomorphism. Two others, the circuit matrix B and the cut-set matrix C, display some important features of the graph and describe the graph only within 2-isomorphism. The path matrix P(x, y) contains even less information than B or C does.

To see further into the structure of the graph, we investigated these matrices, pulled out submatrices A_f, B_f, C_f, I_μ, I_{n-1}, B_t, B_c, C_t, and C_c, and studied them and their interrelationships.

The properties brought out in this chapter do not by any means exhaust the list. Many interesting and useful results are contained in the problems of this chapter:

The converse problem of finding a graph to represent a given matrix has been touched upon lightly in Section 7-5. The problem of realizability, that is, what conditions must a given matrix B satisfy so that a graph can be found whose circuit matrix is B, is very useful and interesting. We shall encounter this problem of realizability again in Chapter 12.

REFERENCES

Some knowledge of elementary matrix algebra was assumed in this chapter. For those not familiar with matrices, dozens of good books are available. Out of these we have listed two, [7-1] and [7-3]. Two somewhat special results, the Binet-Cauchy theorem and Sylvester′s law of nullity, are explained and proved in Appendices A and B, respectively.

Most textbooks referred to earlier have some portion devoted to matrices associated with graphs. Particularly recommended readings are Chapter 13 of Harary [1-5], Chapter 4 of Seshu and Reed [1-13], and Chapter 5 of Busacker and Saaty [1-2]. Gould′s paper [6-4] referred to in the last chapter is also relevant to this chapter. A survey paper by Harary [7-2] proves some of the results given in this chapter.

7-1. AITKEN, A.C, "Determinants and Matrices," 9th ed., Oliver & Boyd Ltd.,

Edinburgh, 1956.

7-2. HARARY, F., "Graphs and Matrices," *SIAM Rev.*, Vol. 9, No. 1, Jan., 1967, 83-90.

7-3. HOHN, F. E., *Elementary Matrix Algebra*, The Macmillan Company, New York, 1958.

PROBLEMS

7-1. Write the incidence matrices for the labeled simple graphs shown in Figs. 1-12 and 4-1 (b). Put the incidence matrix of the graph of Fig. 4-1 (b) in the block-diagonal form of Eq. (7-1).

7-2. Consider the graph in Fig. 4-3. With respect to the spanning tree $\{b, c, e, h, k\}$, write matrices A_f, B_f, and C_f in the forms of Eqs. (7-10), (7-5), and (7-9), respectively. Verify by actual computation Eqs. (7-13) and (7-15).

7-3. Show that for a simple disconnected graph of k components, n vertices, and e edges the ranks of matrices A, B, and C are $n - k$, $e - n + k$, and $n - k$, respectively.

7-4. Label the edges of the graph in Fig. 4-8, and write down its circuit matrix B. Verify observations 1-5 made in Section 7-3 about the properties of matrix B.

7-5. Draw two nonisomorphic, connected, simple, and nonseparable graphs G_1 and G_2, with as small a number of edges as you can, such that the circuit matrices $B(G_1) = B(G_2)$. (*Hint:* G_1 and G_2 are 2-isomorphic, and must be 2-connected.)

7-6. A black box containing a switching network of seven switches−1, 2, 3, 4, 5, 6, and 7−was subjected to the experiment shown in Fig. 7-3. The lamp was lit when each of the following combinations of switches was turned on, in addition to the external switch k, of course: (1, 4, 5), (1,4, 6, 7), (2, 5, 7), (2, 6), (3, 5), and (3, 6, 7). Show the switching network configuration.

7-7. In Section 7-5 a graph was obtained corresponding to a given fundamental circuit matrix. Similarly, sketch a procedure for obtaining a graph if its fundamental cut-set matrix C_f is given. Can you get two different (nonisomorphic) graphs for the same C_f? If yes, how are these different graphs related?

7-8. Show that you can determine a graph within 2-isomorphism if you were given the set of all spanning trees. (*Hint*: From the set of all spanning trees every cut-set can be determined, using Theorem 4-2. And the set of

all cut-sets determines a graph within 2-isomorphism.)

7-9. If the following is the list of all spanning trees of a graph G, determine G.

$\{a, c, d, e\}, \{a, c, d, f\}, \{b, c, d, e\}, \{b, c, d, f\},$
$\{a, c, e, f\}, \{b, c, e, f\}, \{a, d, e, f\}, \{b, d, e, f\},$
$\{a, b, d, e\}, \{a, b, d, f\}, \{a, b, e, f\}.$

7-10. Express the relationship of dualism between two planar, simple graphs in terms of appropriate matrices.

7-11. Characterize simple, self-dual graphs in terms of their circuit and cut-set matrices.

7-12. Prove that

$$\mathbf{B}_f = [\mathbf{I}_\mu \mid \mathbf{A}_c^T \cdot \mathbf{A}_t^{-1T}],$$
$$\mathbf{C}_f = \mathbf{A}_t^{-1} \cdot \mathbf{A}_f.$$

7-13. Write down the path matrix $P(v_1, v_6)$ for the graph in Fig. 4-3. Verify observations 1-4 in Section 7-8 and Theorem 7-7.

7-14. Prove Theorem 7-7.

7-15. Characterize A_f, B_f, C_f, and X matrices of the complete graph of n vertices.

7-16. After having labeled the graph in Fig. 4-8 (as required in Problem 7-4), write its adjacency matrix X. How does the fact that the graph is separable reflect in X? Characterize the adjacency matrix X of a separable graph, in general.

7-17. Complete the proof of Theorem 7-8.

7-18. The diameter of a connected graph is defined (Chapter 3) as the largest distance between two vertices in the graph. Given the adjacency matrix X, how will you determine the diameter of the corresponding graph? (*Hint*: Consider a sum of the powers of X.)

7-19. Show that each diagonal entry in X^3 equals twice the number of triangles passing through the corresponding vertex.

7-20. Prove that the number of spanning trees in a connected graph equals the value of

$$\det(\mathbf{A}_f \cdot \mathbf{A}_f^T),$$

where A_f is the reduced incidence matrix of the graph, and the arithmetic operations are carried out in the real field and *not* mod 2.

7-21. Similar to the circuit or cut-set matrix, define a *spanning-tree matrix* for a connected graph, and observe some of its properties.

7-22. Let C be the cut-set matrix of a nonseparable graph G, and let C(x, y) be the submatrix of C, containing only those rows of C that represent cut-sets with respect to vertices x and y. Show that C(x, y) contains a fundamental cut-set matrix C_f of G.

7-23. For a labeled graph G of n vertices, define an n by n diagonal matrix D (called the *degree matrix* of *G)* such that the ith diagonal entry in D equals the degree of the ith vertex in G. Define another matrix E, obtained from the incidence matrix A of G by arbitrarily replacing one of the two l′s in every column by a −1. Show that if G is a simple, connected graph the following holds (the computations are in the ring of integers and *not* mod 2):

(a) $E \cdot E^T = D - X$.

(b) All cofactors of the matrix D − X are equal.

(c) Each cofactor of D − X equals the number of spanning trees in G, where X is as usual the adjacency matrix of G.

7-24. Use the result obtained in Problem 7-23(c) to prove Cayley′s formula (Theorem 3-10).

7-25. Let x and y be a pair of vertices in a simple nonseparable graph G, and P(x, y) be the corresponding path matrix of G. Prove that every circuit in G is obtained as a mod 2 sum of two rows of P(x, y). From this result, prove that a path matrix in a simple, nonseparable graph determines the graph within 2-isomorphism. *[Hint*: Every circuit Γ in G falls in one of three categories: (1) Γ passes through both x and y; (2) T passes through neither x nor y; or (3) T passes through either x or y. Consider all three cases, and use Theorem 4-11.]

7-26. Prove that for a connected, self-loop-free graph G, subspaces W_Γ and W_s are orthogonal complements of W_G over GF(2) if and only if the number of spanning trees in G is odd. [*Hint*: Define a new e by e matrix $M = \begin{bmatrix} C_f \\ B_f \end{bmatrix}$ Compute det(MM^T), using the identity $C_f B_f^T = B_f C_f^T = 0$, and the Binet-Cauchy theorem (see Appendix A). Show that det(MM^T) = 1 (mod 2) if and only if G has an odd number of spanning trees.]

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† Although matrices are customarily defined over a commutative ring with identity, which need not be a field (such as the ring of integers), we have defined matrix A over a field, GF(2), in keeping with our definition of the vector space W_G in Chapter 6.

† Just as in any two alternative methods of representation, some properties are more evident in one

representation than in the other. For example, the fact that the graph is planar is obvious in Fig. 7-1(a), whereas it is not at all obvious from the matrix in Fig. 7-1(b).

† Some authors (see Busacker and Saaty [1-2], page 109, for example) define x_{ij} as equal to the number of edges incident on both vertices i and j, and thus take into account parallel edges.

† An edge sequence is a sequence of edges in which each edge (except, of course, the first and the last) has one vertex in common with the edge preceding it and one vertex in common with the edge following it. A path, a circuit, and a walk are all special examples of an edge sequence. An edge may appear more than once in an edge sequence.